

ka4.ino

```
1  const int potPin = A0;
2
3  // High-side PMOS control by NMOS
4  const int leftHighCtrl = 2;    // top-left PMOS control
5  const int rightHighCtrl = 3;   // top-right PMOS control
6
7
8  const int leftLow = 5;          // bottom-left NMOS (PWM)
9  const int rightLow = 6;         // bottom-right NMOS (PWM)
10
11
12  const int deadbandLow = 470;
13  const int deadbandHigh = 550;
14
15  void setup() {
16      pinMode(leftHighCtrl, OUTPUT);
17      pinMode(rightHighCtrl, OUTPUT);
18      pinMode(leftLow, OUTPUT);
19      pinMode(rightLow, OUTPUT);
20
21      stopMotor();
22  }
23
24  void loop() {
25      int potValue = analogRead(potPin);
26
27
28      if (potValue >= deadbandLow && potValue <= deadbandHigh) {
29          stopMotor();
30      }
31
32      // Turn one direction
33      else if (potValue > deadbandHigh) {
34          int pwmVal = map(potValue, deadbandHigh + 1, 1023, 0, 255);
35          pwmVal = constrain(pwmVal, 0, 255);
36          driveForward(pwmVal);
37      }
38
39      // Turn the other direction
40      else {
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40     else {
41         int pwmVal = map(potValue, deadbandLow - 1, 0, 0, 255);
42         pwmVal = constrain(pwmVal, 0, 255);
43         driveReverse(pwmVal);
44     }
45
46     delay(5);
47 }
48
49 // -----
50 //   top-left PMOS ON
51 //   bottom-right NMOS PWM
52 //   other diagonal OFF
53 // -----
54 void driveForward(int pwmVal) {
55     // unused
56     digitalWrite(rightHighCtrl, HIGH); // top-right PMOS OFF
57     digitalWrite(leftLow, LOW);        // bottom-left NMOS OFF
58
59     // Turn ON active diagonal
60     digitalWrite(leftHighCtrl, LOW);   // top-left PMOS ON
61     analogWrite(rightLow, pwmVal);     // PWM bottom-right NMOS
62 }
63
64 // -----
65 // Reverse:
66 //   top-right PMOS ON
67 //   bottom-left NMOS PWM
68 //   other diagonal OFF
69 // -----
70 void driveReverse(int pwmVal) {
71     // unused diagonal
72     digitalWrite(leftHighCtrl, HIGH); // top-left PMOS OFF
73     digitalWrite(rightLow, LOW);      // bottom-right NMOS OFF
74
75     // Turn ON active diagonal
76     digitalWrite(rightHighCtrl, LOW); // top-right PMOS ON
77     analogWrite(leftLow, pwmVal);     // PWM bottom-left NMOS

```

```
70 void driveReverse(int pwmVal) {
71     // unused diagonal
72     digitalWrite(leftHighCtrl, HIGH); // top-left PMOS OFF
73     digitalWrite(rightLow, LOW);      // bottom-right NMOS OFF
74
75     // Turn ON active diagonal
76     digitalWrite(rightHighCtrl, LOW); // top-right PMOS ON
77     analogWrite(leftLow, pwmVal);      // PWM bottom-left NMOS
78 }
79
80 // -----
81 // Stop:
82 //   both PMOS OFF
83 //   both low-side NMOS ON
84 // -----
85 void stopMotor() {
86     digitalWrite(leftHighCtrl, HIGH); // PMOS OFF
87     digitalWrite(rightHighCtrl, HIGH); // PMOS OFF
88
89     digitalWrite(leftLow, HIGH);       // low-side ON
90     digitalWrite(rightLow, HIGH);      // low-side ON
91 }
```
